Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B. TECH (MECHANICAL/IP ENGG.) END SEMESTER EXAMINATIONS, NOVEMBER 2019

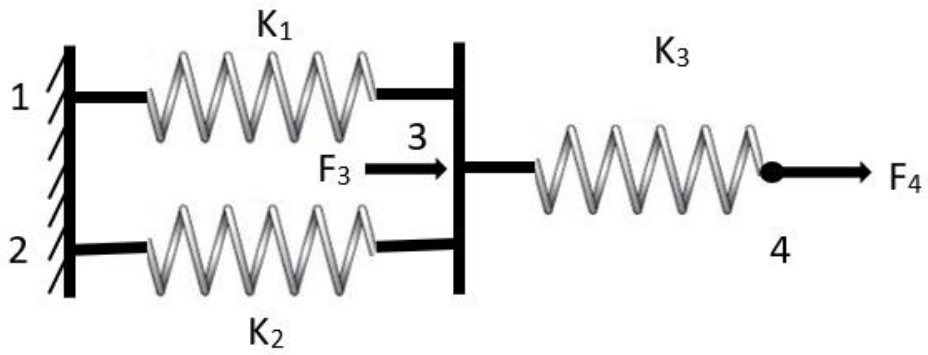
SUBJECT: FINITE ELEMENT METHOD [MME 4102]

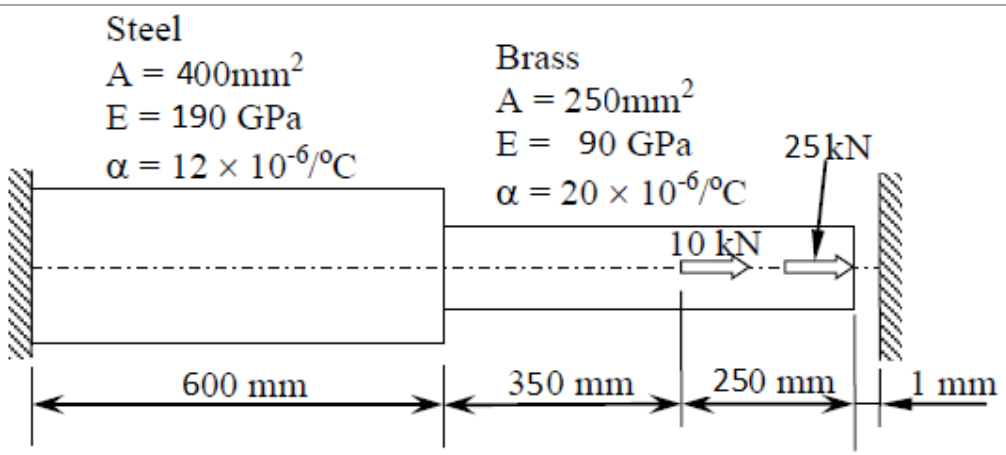
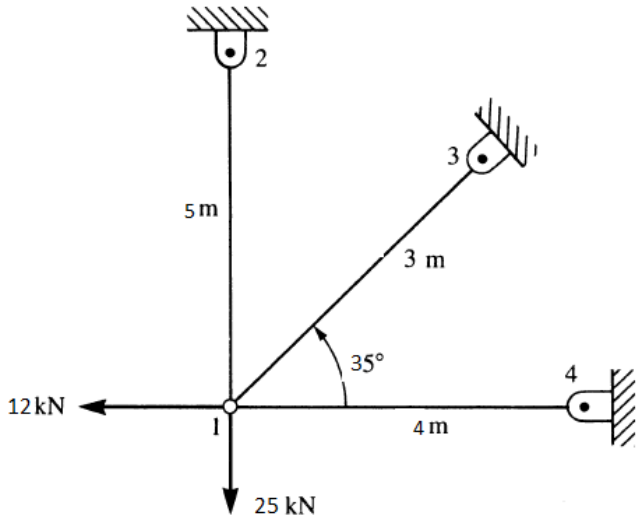
REVISED CREDIT SYSTEM

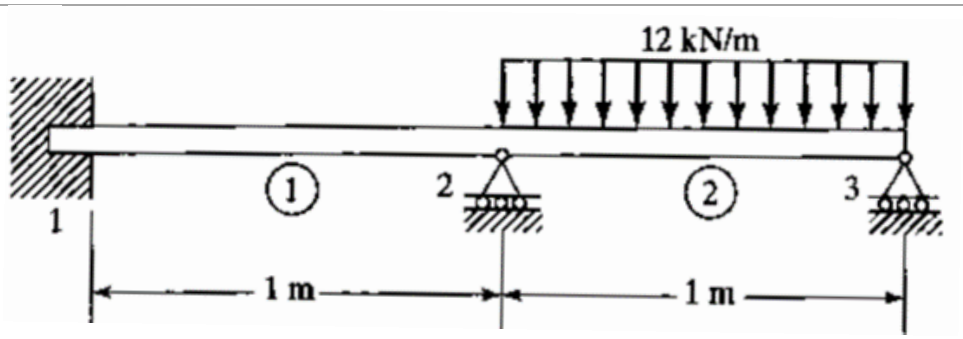
Time: 3 Hours MAX. MARKS: 50

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.
- Draw neat labelled diagrams wherever necessary

1A.	What is a banded matrix and where we come across such matrix in Finite Element Analysis? Explain the Half-Band Width technique used in efficient storage of such matrices in computer memory using a suitable example.	**04M**
1B.	Use Two point Gaussian Quadrature formula to evaluate the integral $$I = \int_0^2 \frac{dx}{\sqrt{4+4x-x^2}}$$	**03M**
1C.	Determine the displacement at nodes for spring system shown **figure Q1C** using principle of minimum potential energy.  Where, K_1=100kN/m, K_2= 2×K_1, K_3= 3×K_2, F_3=0.5 kN, F_4= 2×F_3 **Figure Q1C**	**03M**
2A.	Use finite element method to model the structure shown in **figure Q2A**. Use minimum number of elements. Determine the nodal displacements, element stresses if there is a change in temperature of 40°C. Apply penalty approach of handling boundary conditions.	**05M**

	 **Figure Q2A**	
2B.	Derive the shape functions for 3 noded horizontal 1D element by using a polynomial displacement function.	**03M**
2C.	Mention the general steps followed in FEM?	**02M**
3A.	Determine the horizontal and vertical displacements at the node 1 and stresses in each element for truss configuration shown in **figure Q3A**. Assume Area=4 X 10^{-4} m^2, Young's modulus=210 GPa for all elements.  **Figure Q 3A**	**05M**
3B.	Derive stiffness matrix for 1D thermal element (steady state heat conduction) using direct stiffness method.	**03M**
3C.	Explain what do you mean by specified displacement boundary condition.	**02M**
4A.	Evaluate the element stiffness matrix for a CST element whose details are as follows: (x1, y1) = (55,60), (x2, y2) = (80,95) & (x3, y3) = (25,75). Element thickness = 6 mm, Poisson's Ratio = 0.3, Young's Modulus = 200 GPa. Assume plane stress problem.	**05M**
4B.	Derive the element stiffness matrix for 1D structural (linear) element using Galerkin's weighted residual method.	**05M**
5A.	Determine the slopes at node 2 and 3 for the given beam arrangement as shown in the **figure Q5A**. Assume E = 200 GPa and I = $4x10^6$ mm^4. Note: use minimum no of elements.	**05M**

	 **Figure Q5A**	
5B.	Derive the mass matrix using shape functions for 1D link element.	**03M**
5C.	Explain the following: i) Consistent Mass Matrix. ii) Lumped mass matrix.	**02M**